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Production of Depressive Behaviors in Young Monkeys[1]

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Socially unsophisticated rhesus monkeys ranging in age from 6 to 13 months were individually confined in vertical chambers for a total of 30 days. Upon emergence from the chambers they exhibited significant increases in self-clasping and huddling behaviors and decreases in locomotive and exploratory activity. These behavioral changes were similar in direction to those resulting from separation from attachment objects, which have been described as depressive. The utility of the vertical chamber apparatus as a device for the production and study of depression in monkeys is discussed, and implications for the study of child psychoses noted.

For many years behavioral scientists have attempted to produce psychopathological syndromes in nonhuman subjects by means of experimental techniques (Pavlov, 1927; Liddell, James, & Anderson, 1934; Masserman, 1943). The most successful of these researches employing nonhuman primate subjects have utilized the manipulations of social isolation from birth, either total isolation (Harlow & Harlow, 1962; Mason & Sponholz, 1963; Rowland, 1964) or partial isolation (Cross & Harlow, 1965) as the means for production of behavioral abnormalities. Despite the enormously adversive effect of social isolation from birth upon subsequent social development and the theoretical import of such findings for analysis of the processes of attachment formation and socialization, the direct applicability of this research to problems of human psychopathology seems somewhat limited. This results primarily from the fact that few humans have been raised in total social isolation, and those whose early lives included some degree of partial isolation usually had this experience

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irrevocably confounded with other factors. Extensive research utilizing social isolation of monkey subjects has an additional practical disadvantage. Although a monkey infant develops at approximately four or fives times the rate of its human counterpart, at least 6 months of total isolation initiated at birth is required to produce persisting abnormalities, and this is a relatively lengthy time for both experimental monkeys and human experimentalists.

An alternative approach to production of monkey psychopathology has employed separation from an attachment object as a primary manipulation, and here exists a more direct human parallel. Human infants and children separated from their mothers via hospitalization or institutionalization have been reported to develop "anaclitic depression" (Spitz, 1946; Bowlby, 1960). Young monkeys reared with and then separated from mothers (Seay, Hansen, & Harlow, 1962; Seay & Harlow, 1965; Kaufman & Rosenblum, 1967) or peers (Suomi, Harlow, & Domek, 1970), exhibit a qualitatively similar reaction. In its later stages this reaction is characterized by dramatic declines in active behaviors such as locomotion, exploration, and play, and by increases in self-directed behaviors such as self-clasping and huddling.

Our current interest in the production of monkey depression led Harlow to create an apparatus designed both to intensify reactions to separation and to produce, independent of separation procedures, similar behavioral abnormalities rapidly and consistently in monkey subjects (Harlow, Suomi, & McKinney, 1970). Termed the "vertical chamber," the device is illustrated in Figure 1. It is little more than a stainless steel trough with sides that slope inward to form a rounded bottom. A 3/8 in. wire-mesh floor 1 in. above the bottom of the chamber allows waste material to drop and drain through holes drilled in the stainless steel. The chamber is equipped with a food box and water-bottle holder and is covered by a pyramidal top designed to discourage confined subjects from hanging from the upper part of the trough. A more complete description of the apparatus may be found in Suomi and Harlow (1969).

Depression in humans has been characterized as embodying a state of "helplessness and hopelessness, sunken in a well of despair" (Schmale, 1970). The vertical chamber apparatus was designed on an intuitive basis to reproduce such a psychological well for monkey subjects. An initial study measuring the effects of vertical chamber confinement upon young, socially unsophisticated monkeys was conducted.

METHOD

Subjects

Experimental subjects were four rhesus monkeys: a male 15 months, a male 6 months, a female 13 months, and a female 9 months of age at the beginning of

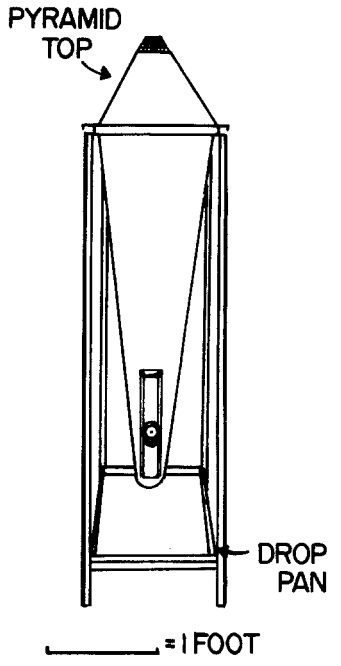


FIG. 1. Diagram of vertical chamber apparatus.

the experiment. All had been separated from their mothers at birth and reared for the first 30 days of life in the laboratory nursery according to the procedure described by Blomquist and Harlow (1961). Subsequently they were individually housed in bare 3/4-in. wire-mesh cages 2.5 x 2 x 2.5 ft in. size, where they could see and hear but not physically contact other monkeys. The animals remained undisturbed in their individual cages until the start of the experiment.

Twelve rhesus monkeys, six males and six females ranging in age from 6 to 24 months, served as control animals. They had been raised in the laboratory according to the same procedure employed for the experimental subjects, having been separated from their mothers at birth, reared in the nursery for the first 30 days of life, and subsequently confined to individual cages 2.5 x 2 x 2.5 ft in. size.

Procedure

Each experimental subject was removed from its living cage and placed in a vertical chamber for 20 days, then returned to its home cage for a period of 7 days, placed in the chamber for an additional 10-day period, and finally returned to its home cage. Thus the total duration of chamber confinement for each

experimental subject was 30 days. Control subjects remained in their home cages throughout the experiment.

Data Collection

Two daily 5-minute observations of each experimental subject in its home cage were initiated 14 days prior to the first chamber confinement in order to assess prechamber behavioral baselines. During each observational session behaviors falling into each of four categories were noted for presence or absence during each of the twenty 15-second intervals which comprised the 5-minute session. A description of the behaviors encompassed by each category is provided in Table 1.

Table 1

Description of Behavioral Categories Employed

Self-clasp	Huddle	Locomotion	Environmental exploration
Clasping of any part of own body with hand(s) and/or foot (feet).	Self-enclosed, fetal-like position incorporating any or all patterns of self-clasp, self-embrace, or lowered head, all of which may be accompanied by repetitive, non-locomotive forward and backward movement.	Ambulation of one or more full steps.	Tactual and/or manipulation by subject of inanimate objects, e.g., towel, toy, cage lock, etc.

The behaviors of the experimental subjects were not formally scored during the two periods of chamber confinement. However, twice-daily observational sessions were reinstated during the 7-day period between incarcerations and were continued for 8 weeks after each subject had been returned to its home cage following the second removal from the vertical chamber.

Each control subject was observed in its home cage for five 5-minute sessions over a period of 3 weeks. The categories employed for scoring experimental subjects were included in the observations of the control subjects.

All observations were made between the hours of 9:00 a.m. and 4:30 p.m. Three testers, previously trained to a rigorous laboratory reliability criterion, recorded the subjects' behaviors, each scoring approximately one third of the sessions. Order of testing among subjects was randomly determined.

Data Analysis

Scores of the experimental subjects for each category were summed over six time-blocks: the 14 days prior to chamber confinement, the 7 days between the first 20 and final 10 days of chamber confinement, and each of the 2-week periods following the second removal from chamber confinement. For each of these sums a mean was calculated representing the average number of 15-second intervals per 5-minute observational session during which subject behaviors encompassed by the category were observed. These means were subjected to a one-way repeated measures analysis of variance, with time-block as the repeated measure, for each category. A Duncan New Multiple Range Test (Duncan, 1955) was subsequently performed upon the scores of those categories whose analysis had yielded a significant time block effect.

In addition, for every control subject a mean was calculated for each category over the five observation sessions, representing the average number of 15-second intervals per 5-minute observational session during which behaviors encompassed by the category were observed. These means were then compared by two-tailed *t*-tests with the category means of the experimental subjects during the 2-week time block prior to chamber confinement, and with comparable experimental subject means during the final 2-week time block following chamber confinement.

RESULTS

The results of this study strongly indicated that the behaviors of the experimental subjects were significantly altered, as shown in Figure 2, following vertical chamber confinement. Postincarceration levels of self-clasping and huddling were far higher than baseline levels, and these elevations were maintained throughout follow-up testing. Correspondingly the observed levels of locomotive and exploratory behaviors dropped sharply following subject chamber confinement and failed to return to baseline. The means of all categories recorded for the experimental subjects during the final 2 weeks of testing, 2 months following removal from the vertical chambers, lay outside the statistically acceptable range of behavioral levels exhibited by the control monkeys, despite the fact that the experimental animals' baseline levels fell within such limits.

Analysis of variance of the experimental subjects' scores yielded significant time block effects for every category ($F = 4.32$, $df = 5/15$, $p < .025$; $F = 15.04$, $df = 5/15$, $p < .0005$; $F = 4.18$, $df = 5/15$, $p < .025$; and $F = 4.73$, $df = 5/15$, $p < .01$ for self-clasp, huddle, locomotion, and environmental exploration,

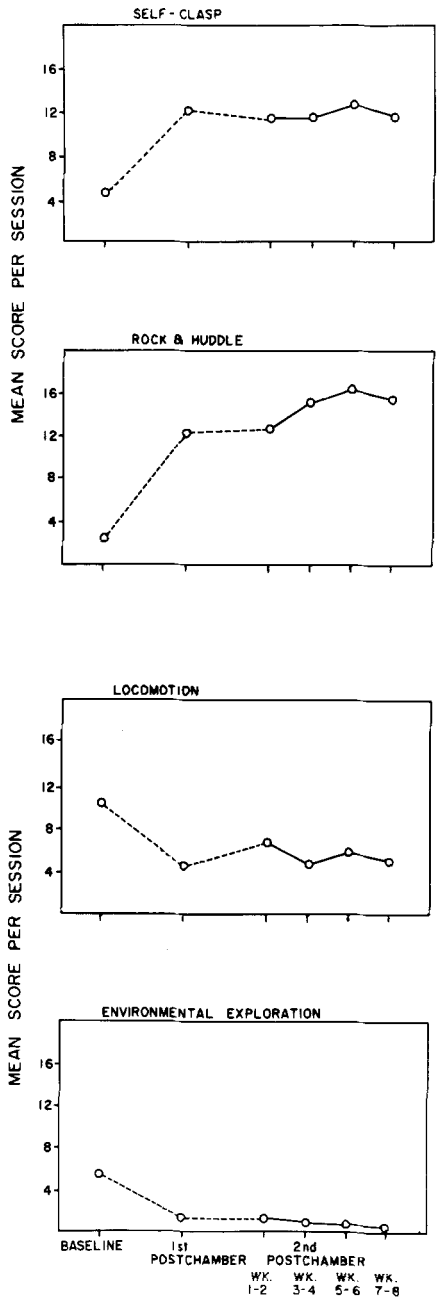


FIG. 2. Observed behavioral changes.

respectively). Subsequent tests indicated that for each category, the baseline levels differed significantly from those of all other time blocks (all p's $< .01$), except for the locomotion analysis where two of the four comparisons yielded differences significant at "only" the .05 level. No other significant time-block differences were detected. Thus, the alterations from baseline levels observed in the behaviors of the experimental subjects immediately following confinement remained stable for the duration of the follow-up testing period.

The behavioral levels of the experimental subjects prior to confinement did not differ significantly from those of the control population. However, significant deviations by the experimental subjects from the control means were detected in all categories 2 months after confinement had been completed. These results are summarized in Table 2. The experimental subjects, once indistinguishable from a group of monkeys of comparable age and rearing experience, exhibited profound behavioral anomalies following a total of 30 days confinement in vertical chambers.

Table 2

Comparisons between Control and Experimental Subjects

Behavioral category	Experimental baseline vs. control		Final experimental time block vs. control	
	$t_{(14)}$	p	$t_{(14)}$	p
Self-clasp	1.97	<.10	4.36	<.01
Huddle	0.52		8.72	<.01
Locomotion	0.62		3.91	<.01
Environmental exploration	0.29		3.39	<.01

Discussion

The results of this experiment indicated that pervasive behavioral changes of a profound nature could be rapidly produced via vertical chamber confinement and that the altered levels of behavior persisted long after the subjects had been removed from the chambers. The significance of these findings is twofold.

First, it is clear that the vertical chamber has enormous potential for rapid production of psychopathological behavior in monkeys. Total isolation of 3 months duration results in only transient behavioral disturbance (Boelkins, 1963; Griffin & Harlow, 1966). Furthermore, the effectiveness of isolation decreases considerably if not initiated at or shortly after birth (Rowland, 1964;

Clark, 1968). In contrast, vertical chamber confinement of only 30 days duration produced profound disturbances even in subjects over a year of age at time of incarceration. Unfortunately, two of the experimental subjects were employed in another study immediately after formal testing had been completed. However, the other two chambered monkeys remained in their home cages for almost a year, and informal observations indicated little, if any, recovery by these animals. Clearly, the abnormalities produced by vertical chamber confinement were far from transient.

Significant also is the nature of the observed abnormalities. The changes in behavior exhibited by the chambered subjects—increases in self-clasping and huddling and decreases in locomotion and exploration—were virtually identical to behavioral changes observed in monkeys separated from objects of attachment, changes which have been termed "depressive" by some authors (Kaufman & Rosenblum, 1967). This is a remarkable finding, in that vertical chamber confinement apparently can produce among socially unsophisticated subjects behavioral components similar to those produced by a social manipulation among well-socialized monkeys.

It is unfortunate that formal observation of the subjects was not maintained during the periods of confinement. However, informal observations indicated that when monkeys were first placed in the chambers they exhibited a great deal of activity. Within a few days, however, most subjects had assumed a huddling posture in a corner of the chamber, as illustrated in Figure 3. While we did not ask our subjects if they felt helpless and/or hopeless, their posture and lack of activity seemed to indicate an attitude of "giving up." Seldom has intuition been proven so vigorously validated.

It is not clear what aspect of vertical chamber confinement was responsible for subsequent subject behaviors. Researches investigating parameters such as chamber shape, chamber construction, duration of confinement, age at time of confinement, and prior social experience have been initiated to answer this question. Nor is it clear precisely what is the relationship, if any, between chamber confinement and separation which produces similar behavioral consequences. Current research manipulating social parameters in combination with chamber confinement may uncover important findings in this area. However, it is clear that vertical chamber confinement can produce profound and persistent behavioral abnormality within a relatively short time period, and it is probable that this is mediated not by one, but many, variables.

The possible relationship of the abnormal behaviors exhibited by these monkeys to forms of human psychopathology will be discussed in a separate

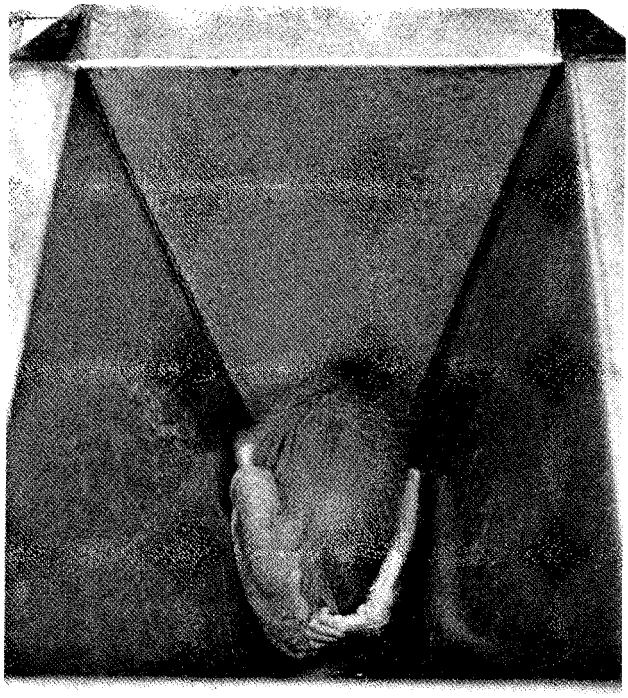


FIG. 3. Typical subject posture within vertical chamber.

paper.[3] The theoretical linkages between monkey and human psychopathology, including psychoses in childhood, merit exploration since there is great potential in the establishment of primate models for human behavioral disorders. Such models certainly have heuristic value and even possibly practical therapeutic value. However, appropriate caution must be exercised in the creation of such model systems and the interpretation of resulting data to avoid premature closure in a newly developing area.

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